

Finding the unknown using multiple protein test

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section 6, Name of experiment: Finding unknown solutions using protein testing.

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Introduction:

Proteins are essential macromolecules that play a wide range of structural and functional roles within living organisms (Rashid Abu Alia, 2025). They are composed of amino acids connected by peptide bonds (Suhail Al-Khatib, 2025). In this experiment, several tests were performed to identify the proteins and amino acids in different unknown samples (Rashid Abu Alia, 2025). The tests used were the Biuret test, the Ninhydrin test, and the Xanthoproteic test (Rashid Abu Alia, 2025). By comparing the color changes and results obtained from each test, we aimed to determine which unknown sample contained protein or amino acids and thus identify the composition of each beaker (Rashid Abu Alia, 2025).

Objective:

The aim of the experiment is to identify several unknown solutions using chemical tests (Biuret test, the Ninhydrin test, and the Xanthoproteic test).

Materials and tools:

Tube, tube holder, gloves, several droppers, a few of unknown solutions, a device that preserves volatile (fast evaporating) materials, biuret reagent, Ninhydrin solution, Concentrated HNO₃.

Method:

First test: Biuret

1-We brought seven tubes and put 1 ml of the unknown substances in them, arranged in alphabetical order, and we put distilled water in the eighth tube.

2-We added 1 ml of NaOH, then we added 5 drops of Biuret reagent, and then we noticed a color change.

*See Figure 1

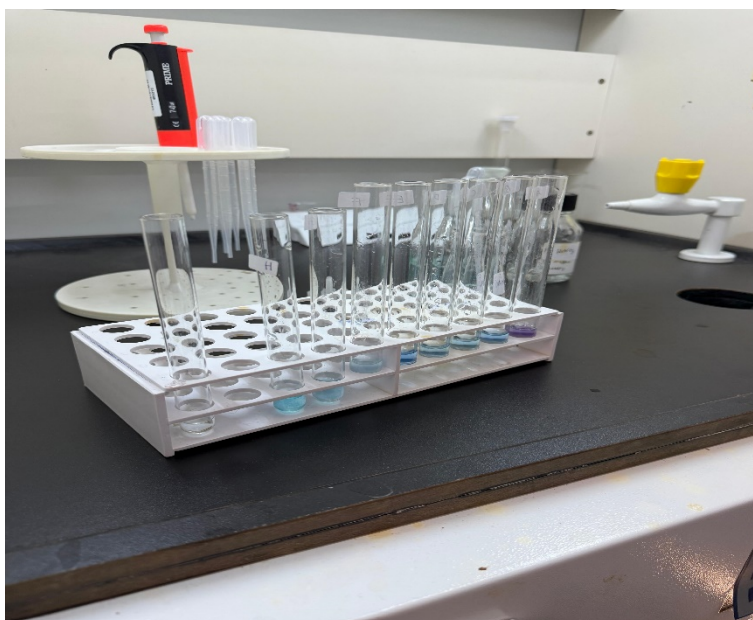


Figure 1

Arrange the tubes from right to left **A-H**. From the figure, we notice that tube **A** is positive and purple, while the rest of the tubes are negative and blue.

Second test: Ninhydrin

- 1-We added 1 ml of unknown solutions to the tubes.
- 2-We named the tubes alphabetically A-H.
- 3-We added 2 ml of Ninhydrin solution.
- 4-We Heated the mixture in water bath for 2 min.
- 5-After removing it from the water bath, we noticed a change in color.

***See Figure 2**

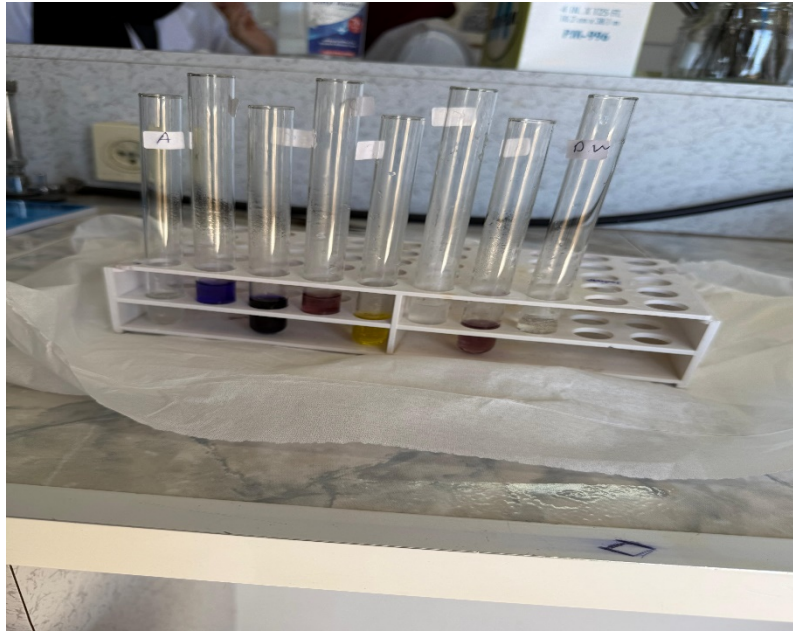


Figure 2

The tubes are arranged from left to right in alphabetical order **A-H**. We notice the change in color. The positive solutions are yellow or purple. The remaining solutions are negative.

Third test: Xanthoproteic

- 1-We added 1 ml of unknown solutions to the tubes.
- 2-We named the tubes alphabetically A-H.
- 3-Then we added 0.5 ml of concentrated HNO_3 To solutions.
- 4-Then we put the tubes into the water bath for 2 min.
- 5-After the time was up, we took it out and noticed the change in colors.

***See Figure 3**

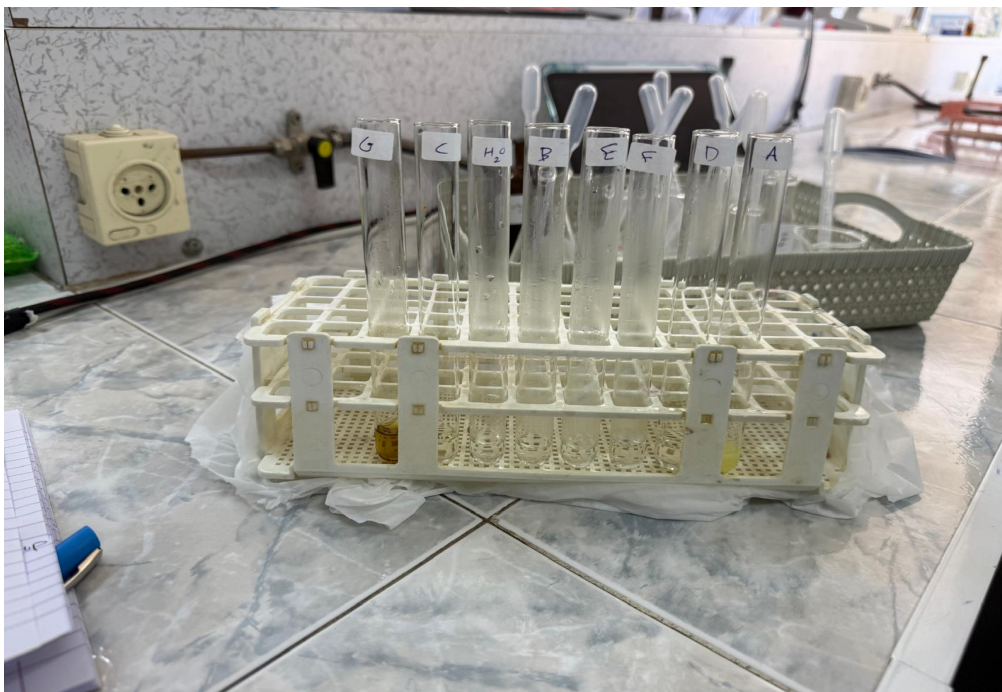


Figure 3

In this figure, the tubes are not arranged in order, but I have labeled them as shown in the figure.

As we have noticed, the transparent color indicates a negative solution, while the rest of the colors indicate a positive one.

Results:

Objective: The aim of the experiment is to identify several unknown solutions using chemical tests (**Biuret test, the Ninhydrin test, and the Xanthoproteic test**).

We did this experiment to identify different unknown solutions. In the **first test**, we used the biuret test to detect protein in the solutions if it was present. The test contains copper sulfate, so the color of the solution changes to purple if protein is present. **As we can see from Table 1 and Figure 1.**

In the **second test**, we used the Ninhydrin test to detect the presence of amino acids in the unknown solutions. Ninhydrin reacts with the amino group of amino acids, producing a purple or blue color. If no free amino acids are present, the solution remains colorless or shows only a faint yellow color. As shown in **Table 1 and Figure 2**, it indicates a positive result for amino acids, while the other samples remained negative.

In the **third test**, we performed the Xanthoproteic test to identify aromatic amino acids. When concentrated nitric acid is added to a protein solution, it reacts with the aromatic rings, producing a yellow color that turns orange upon the addition of an alkali. As observed in **Table 1 and Figure 3**, **tube G** turned yellow after heating, confirming the presence of aromatic amino acids, while all other beakers showed no color change, indicating negative results.

Unknown Solutions	Biuret	Ninhydrin	Xanthoproteic	Identity
A	+Violet	-Colorless	-Colorless	Albumin
B	-Blue	+Purple	-Colorless	Glycine/Alanine/Phenylalanine
C	-Blue	+Purple	-Colorless	Glycine/Alanine/Phenylalanine
D	-Blue	+Purple	-Colorless	Glycine/Alanine/Phenylalanine
E	-Blue	+yellow	-Colorless	Proline
F	-Blue	-Colorless	-Colorless	Sturch
G	-Blue	+Purple	+Orange	Tyrosine
H2O	-Blue	-Colorless	-Colorless	H2O

Table 1

Conclusion:

By performing the first test, we determined that tube **A** contains a positive solution, and we were able to identify the solution, which is: **Albumin**.

Then we did the second test to identify the two tubes **E** and **F** by changing their color, which are: **Proline and Sturch** respectively.

As for the third test, we were able to identify one solution from the tubes **G**, which is: **Tyrosine**.

Unknown Solutions	Biuret	Ninhydrin	Xanthoproteic	Identity
A	+Violet	-Colorless	-Colorless	Albumin
B	-Blue	+Purple	-Colorless	Glycine/Alanine/Phenylalanine
C	-Blue	+Purple	-Colorless	Glycine/Alanine/Phenylalanine
D	-Blue	+Purple	-Colorless	Glycine/Alanine/Phenylalanine
E	-Blue	+yellow	-Colorless	Proline
F	-Blue	-Colorless	-Colorless	Sturch
G	-Blue	+Purple	+Orange	Tyrosine
H2O	-Blue	-Colorless	-Colorless	H2O

Table 1

References:

- 1-CAMPBELL BIOLOGY Chapter 5.
- 2-Rashid Abu Alia: Doctor teaching general biology at Hebron University.
- 3-Suhail Al-Khatib: Doctor teaching general biology at Hebron University.
- 4-The first figure shows the results of the first test.
- 5-Figure 2 is an image showing the results of the second test.
- 6-Figure 3 is an image showing the results of the third test.

7-Table 1 shows the results of all tests, along with the identity and colors of the solutions.